

SmartHospital Analytics Report

Fajar Faisal

Ontario Tech University

BUSI 4590U: Topics in Technology Management: Data Analytics

Date: August 5th 2026

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Executive Summary

The SmartHospital performed an analytics case study that aimed at analyzing the process of patient flow and service efficiency in clinical departments. The main purpose of this case study was the identification of the reasons that determine the patient wait time and the length of stay through combining information from operational, lab, physicians and department data into one analytical database. Long wait times and lengthy patient admissions can negatively affect patient satisfaction, resource use, and overall healthcare effectiveness, highlighting the need to identify the reasons for these delays.

To achieve this objective, four databases were imported to SAS, cleaned and merged by using the same unique key fields. Additional analytical variables were developed for the purposes of analysis such as day-of-week indicator, age group, patient priority level and long wait time. Different types of statistical methods were used in the case study including descriptive statistics, graphs, hypothesis tests, ANOVA, correlation analysis and regression analysis.

The study investigated if average wait times aligned with management standards, if wait times varied among departments and shifts, and which elements had the greatest impact on patient duration of stay. The results uncovered significant operational trends and emphasized aspects where efficiency might be enhanced. Drawing from the statistical data, a number of suggestions were created to minimize delays, optimize resource distribution, and enhance patient treatment. Overall, this study demonstrates how data analytics can support evidence-based decision making and improve healthcare service performance.

Case Background and Objectives

Healthcare organizations are under increasing pressure to ensure the quality of care while also reducing costs and being efficient. Long waiting lists and patient stays not only risk the patient experience but also the hospital's performance and reputation. As a result, it is critical for hospitals to constantly evaluate their performance in order to remain competitive and meet the expectations of the belligerent market.

Data analytics can benefit healthcare organizations by providing critical insights that allow the entities to make educated decisions and remain efficient.

The company SmartHospital has requested an analytical audit of its operations to assess the functioning of their eight clinical departments. Management raised concerns about the organization's ability to provide services in line with its standards and the causes of prolonged treatments. The analysis processed patient visits, laboratory results, physicians and departments data, and statistical analysis. This paper will discuss the rationale behind the analytical procedures and provide an assessment.

Project Objectives

The main objectives of this study were:

- Integrate several databases of the hospital into one analytic database.
- Assess the hospital's waiting time and length of stay.
- Compare the two indicators based on departments and shifts.
- Examine the factors associated with the patient's length of stay.
- Establish whether the waiting time is above or below the set thresholds by the management.
- Interpret the effect of variables on the two indicators.
- Formulate recommendations based on the evidence.

By accomplishing the listed objectives, SmartHospital will have a clear understanding of the influencers of the patient's experience and have an opportunity to improve its performance.

Data and Variables

This research used four datasets from SmartHospital to examine patient movement and operational effectiveness. The datasets comprised Patient Visits, Laboratory Outcomes, Physicians, and Departments. These datasets were combined utilizing shared identifiers, such as Visit_ID, Physician_ID, and Department_ID, to form an all-encompassing analytical database. Table 1 presents an overview of the datasets utilized in the research along with their specific purposes.

Dataset	Primary Key	Description
SmartHospital_Patient_Visits	Patient_Visit_ID	Patient demographics, wait times, treatment times, triage levels, and length of stay.
SmartHospital_Lab_Results	Visit_ID	Laboratory testing and processing information
SmartHospital_Physicians	Physician_ID	Physician characteristics and experience data.
SmartHospital_Departments	Department_ID	Department information and operational characteristics.

Extra analytical variables were generated in SAS through conditional logic and date/time functions to assist with patient classification and operational analysis.

Analytical Variables

Variable	Description
Weekend_Flag	Identifies weekday versus weekend visits
Arrival_Hour	Extracts patient arrival hour
Age_Group	Categorizes patients by age range
High_Priority_Flag	Identifies urgent triage cases
Long_Wait_Flag	Identifies visits exceeding wait-time benchmarks
Weekday	Determines the day of patient arrival

The comprehensive combined dataset contained 3,000 patient visits and 37 variables, offering an overall perspective of hospital operations. This dataset was utilized for descriptive analysis, testing hypotheses, ANOVA, and regression modeling.

SAS Methodology

The analysis was conducted in SAS, and the process started with importing the data and ended with conducting the analysis. The data was imported in SAS by using PROC IMPORT. In total there were

four excel files. All files were imported into a permanently stored SAS library. The properties of the files using PROC CONTENTS, and we made sure that the data types for each variable are similar. In the data set, Particular attention to date and datetime variables, such as Arrival_Date and Arrival_DateTime.

Data prepared the data to analyze by creating new variables in SAS using the condition and date functions. As a result, we created the following variables: Weekday, Weekend_Flag, Arrival_Hour, Age_Group, High_Priority_Flag, Long_Wait_Flag to make further operational comparisons. Then, merged the four data sets using PROC SQL. The visit dataset was merged with lab results using the Visit_ID , physician information using the Physician_ID , and the department information using the Department_ID . As a result, the HOSPITAL_ANALYSIS data set consists of 3000 hospital visits and 37 variables.

Descriptive analysis using PROC PRINT, PROC FREQ, PROC MEANS, PROC TABULATE, PROC SQL. In addition, built several graphical summaries such as histograms, bar charts, boxplots, scatter plots, and regression plots. Moreover, we applied several statistical tests including one sample t-test, independent sample t-test, correlation analysis, and linear regression analysis to explore the relationships between variables and to make predictions.

Descriptive and Graphical Analysis

Descriptive statistics and graphical analysis were performed to give a summary of SmartHospital's operations and patient movement trends. The combined dataset included 3,000 patient visits across eight clinical departments, facilitating a thorough assessment of hospital performance. Summary statistics, frequency distributions, and graphical representations were utilized to analyze patient attributes, wait durations, length of hospitalization, departmental activities, triage levels, and shift effectiveness.

Key Operational Statistics

HOSPITAL WAIT TIME SUMMARY:

The MEANS Procedure

Variable	Label	Mean	Median	Minimum	Maximum	Std Dev
Age	Age	42.752000	44.000000	1.000000	92.000000	23.9809826
Wait_Time_Minutes	Wait_Time_Minutes	42.956000	43.000000	6.000000	98.000000	17.5462823

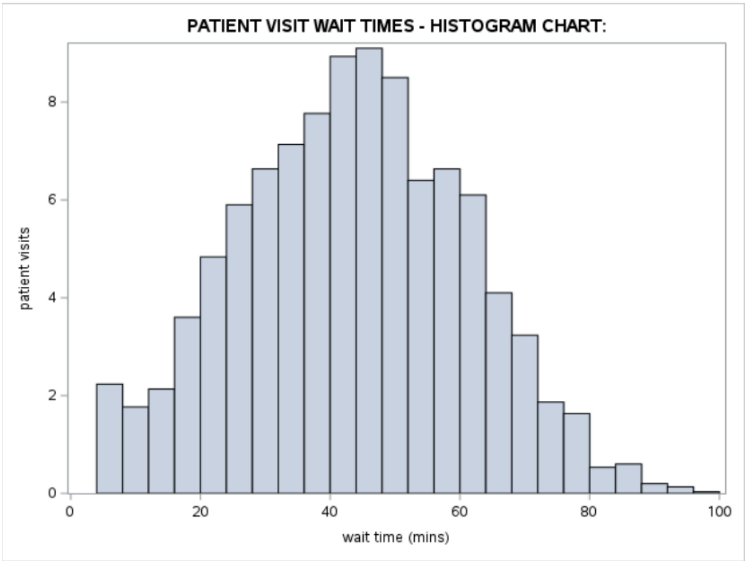


Program Summary - Group2_Project.sas

Variable	Label	Mean	Median	Minimum	Maximum	Std Dev
Treatment_Time_Minutes	Treatment_Time_Minutes	88.814000	89.000000	28.000000	157.000000	19.2655078
Total_Length_of_Stay_Minutes	Total_Length_of_Stay_Minutes	188.859000	187.000000	86.000000	301.000000	40.2432722
Lab_Delay_Minutes	Lab_Delay_Minutes	24.9876667	25.000000	8.000000	48.000000	6.5501004
Number_of_Tests	Number_of_Tests	3.003000	3.000000	1.000000	5.000000	0.8305947

The average waiting time for patients was 42.96 minutes, which is slightly below the hospital’s established standard of 45 minutes. The average total length of stay was 188.86 minutes, meaning that patients spend over three hours at the hospital from arrival until discharge or admission. The significant standard deviation indicates that there is a great variance in the patients’ experiences, which implies a possibility to improve the process.

Distribution of Patient Wait Times



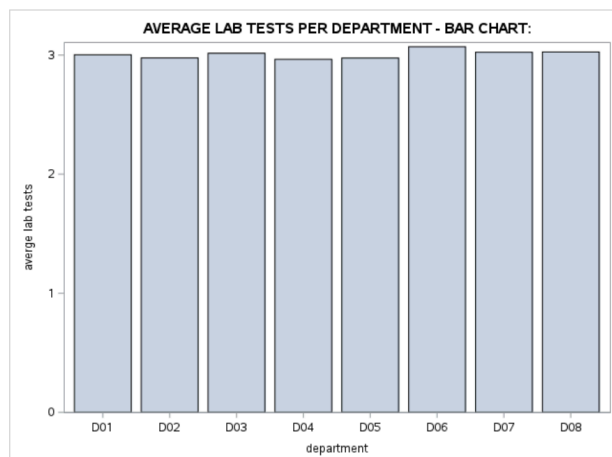
The histogram below represents the patient's wait time at all SmartHospital's visits. From the graph, one can see that the most common wait time is around the mean, which comprises about 43 minutes. Still, the graph also indicates that patients could sometimes wait almost 98 minutes to be served. Therefore, the process has significant variability in wait time, which might be related to certain difficulties in the workflow.

Department Patient Volume

Department_ID		
Department_ID	Frequency	Percent
D01	780	26.00
D02	328	10.93
D03	334	11.13
D04	280	9.33
D05	515	17.17
D06	315	10.50
D07	254	8.47
D08	194	6.47

The highest percentage of calls was received by department D01 – 26.0%. At the same time, only 6.47% of visits were handled by the D08 department. Therefore, it can be concluded that the number of patients' visits is unequally distributed among the departments. It might cause a strain on the workflow in the high-load departments and, consequently, longer waits for treatment.

Average Number of Lab Tests per Department



The bar chart given below represents the average number of laboratory tests performed in various hospital departments. The difference in numbers is associated with the variations in the demands of laboratory tests among different departments. It can be assumed that the increased activity of laboratory tests in different units may require greater staffing and additional laboratory resources to provide proper treatment for patients.

Triage, Disposition, and Arrival Mode Summary ([Figure 1](#))

Triage Level Distribution

Triage_Level				
Triage_Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	241	8.03	241	8.03
2	536	17.87	777	25.90
3	1048	34.93	1825	60.83
4	840	28.00	2665	88.83
5	335	11.17	3000	100.00

According to the chart, the number of visits of triage level 3 was the highest among other levels and amounted to 1048 or 34.93%. It was followed by the fourth level, which was 840 or 28%. Thirty-three percent of customers got a level 5 which is the lowest one (11.17%). The number of patients with the highest priority (level 1) was 241 that comprised 8.03% of cases. The data can be helpful in determining how many healthcare workers should be employed and what resources should be allocated on a priority basis.

Disposition Distribution ([Figure 2](#))

Disposition				
Disposition	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Admitted	1476	49.20	1476	49.20
Discharged	1315	43.83	2791	93.03
Transferred	209	6.97	3000	100.00

The disposition analysis indicates that a total of 1,476 patients, or 49.20%, were admitted to the hospital.

This outcome was reported as the most common one. Meanwhile, 1,315 patients, or 43.83%, were discharged, and 209 patients, or 6.97%, were transferred to another facility or service. The high admission rate reflects that more than half of the patients require further treatment and, therefore, increases the pressure on the inpatient services. It is essential for hospital administration to pay attention to this issue when determining the number of workforce and beds available.

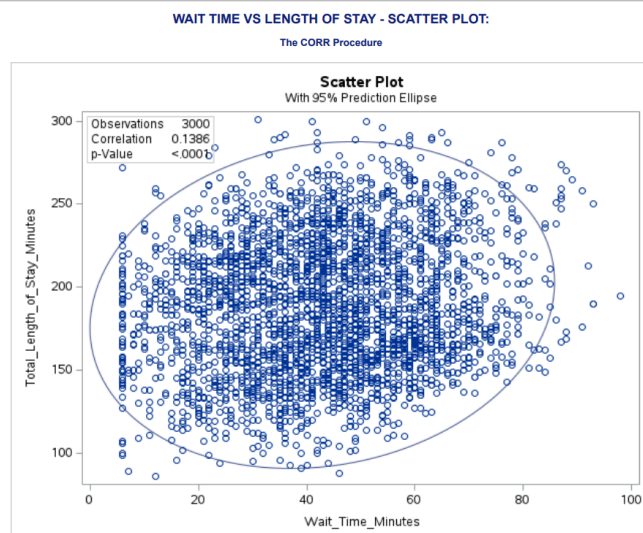
Average Wait Time by Shift

AVERAGE WAIT TIME PER SHIFTS SUMMARY:

Shift	Wait_Time_Minutes
	Mean
Afternoon	45.49
Morning	35.04
Night	52.17

Overall, the average waiting time differed significantly from one shift to another. Specifically, patients who visited the facility in the morning recorded the shortest average waiting time, which was 35.04 minutes, compared to those who arrived in the night, whose average waiting time was the longest, 52.17 minutes. Therefore, the difference in average waiting time between morning and night shifts is equal to approximately 17 minutes, which indicates a possibility of a connection between the number of staff members and patient flow.

Wait Time vs. Length of Stay [\(Figure 3\)](#)



The scatter plot presents the information on the relationship between the patient's wait time and the length of stay. The Pearson's correlation coefficient for the two variables, Wait_Time_Minutes and Total_Length_of_Stay_Minutes, was $r = 0.1386$ ($p < 0.0001$). Thus, it can be concluded that there exists a statistically significant, but small, positive correlation between the two variables. The results suggest that patients who enter the system at a later point tend to stay for a slightly longer time, but the wait time is not the primary factor influencing the length of stay. Other variables, for example, patients' conditions and the workload of the system, may contribute greatly to the duration of the hospital visit.

Overall, the descriptive and graphical analysis identified several trends in the operations of the hospital. Specifically, the trends were identified within different departments, including the different patient groups and shifts. The trends in patient volume, wait time, and overall outcome are critical in understanding the inferential statistics discussed in the next section.

Hypothesis Tests

Two separate hypothesis tests were done in order to assess the hospital's waiting time. First, a one-sample t-test was done in order to determine the claim that the average wait time is 45 minutes as stated by the management's standard. The second test was done to compare the waiting time during weekends and weekdays using an independent two-sample t-test.

One-Sample t-test: Average Wait Time vs. 45 minute Benchmark [\(Figure 4\)](#)

$H_0: \mu = 45$ minutes and $H_1: \mu \neq 45$ minutes

The process involved analyzing 3,000 patient visits. The average waiting time is 42.96 with a standard deviation of 17.55 minutes. The test result is therefore: $t = -6.38$ and $p = 0.0001$.

Since the p-value (0.0001) is less than the alpha (0.05), the null hypothesis was rejected because the calculated value ($t = -6.38$) is beyond the critical value ($t = 1.962$). Thus, the results suggest that

SmartHospital's average patient wait time is different from the set mean of 45 minutes. Since the calculated (sample) mean is less than the set (population) mean, it can be concluded that the SmartHospital average patient's wait time was better than the set management expectation during the studied period.

Independent Two-Sample t-Test: Weekday vs. Weekend Wait Times [\(Figure 5\)](#)

The average waiting time on weekdays was 41.44 minutes while that on weekends was 46.45 minutes. The equality of variances test revealed that the variances were equal with a p-value of 0.8890. As such, the pooled t-test was performed.

The pooled t-test revealed that there was a significant difference in waiting time ($t = -7.25$, $p < 0.0001$). The results showed that patients waited significantly longer on weekends than on weekdays. Specifically, patients waited almost five minutes longer on average on weekends compared to weekdays. This might be an indication of high demand for services on weekends.

ANOVA and Post Hoc Results

One-Way ANOVA Results [\(Figure 6\)](#)

A one-way ANOVA was performed to test whether the patient wait time varied across the eight hospital departments. The outcome variable of the ANOVA was the Wait_Time_Minutes, and the sample size was 3000 patient visits.

Hypotheses of the test were as follows:

- H_0 : The mean patient wait time is equal across all departments.
- H_1 : The mean patient wait time differs across at least one department.

The ANOVA results revealed that wait times differed significantly across departments $F(7, 2992) = 45.66$, $p < 0.0001$; therefore, the researchers rejected the null hypothesis. The result implies that patient wait time varied significantly across the eight departments. Specifically, the one-way ANOVA demonstrated

that approximately 9.65% of variance in patient wait time can be predicted by the department waiting time, $R^2 = 0.0965$, which suggests that the department significantly predicts wait time.

Post Hoc Analysis (shift) [\(Figure 7\)](#)

Since the shift was statically significant ($p < 0.0001$), the post-hoc multiple comparison test was completed with Turkey Kramer analysis to reveal which shifts differed from each other in terms of waiting time. All three shifts were found to be statistically different from each other ($p < 0.0001$). Specifically, the Night shift waiting time was the highest with (50.18 minutes), and Morning shift waiting time was the shortest with (34.75 minutes). Thus, the Afternoon shift waiting time of (43.98 minutes) was in the middle.

The waiting time for Morning was significantly shorter than for the Night shift with a mean difference of 15.42 minutes. Moreover, the Afternoon waiting time was significantly higher than Morning waiting time with a mean difference of 9.23 minutes. Finally, the Night shift waiting time was significantly higher than the Afternoon waiting time with a mean difference of 6.19 minutes. In this case, the waiting time for patients varies per shift with the shortest waiting time observed in the Morning and the longest waiting time observed in the Night. Therefore, it is recommended to consider adding more resources in the Night shift, as the patient load may not be appropriately covered by the staff in that shift, unlike in the Morning.

Correlation and Regression Results

Correlation

Pearson Correlation Analysis of Patient Flow Metrics							
The CORR Procedure							
4 Variables: Wait_Time_Minutes Treatment_Time_Minutes Total_Length_of_Stay_Minutes Number_of_Tests							
Simple Statistics							
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
Wait_Time_Minutes	3000	42.95600	17.54628	128868	6.00000	98.00000	Wait_Time_Minutes
Treatment_Time_Minutes	3000	88.81400	19.26551	266442	28.00000	157.00000	Treatment_Time_Minutes
Total_Length_of_Stay_Minutes	3000	188.85900	40.24327	566577	86.00000	301.00000	Total_Length_of_Stay_Minutes
Number_of_Tests	3000	3.00300	0.83059	9009	1.00000	5.00000	Number_of_Tests

Pearson Correlation Coefficients, N = 3000					
Prob > r under H0: Rho=0					
	Wait_Time_Minutes	Treatment_Time_Minutes	Total_Length_of_Stay_Minutes	Number_of_Tests	
Wait_Time_Minutes	1.00000	-0.34204 <.0001	0.13863 <.0001	-0.52734 <.0001	
Treatment_Time_Minutes	-0.34204 <.0001	1.00000	0.51486 <.0001	0.54630 <.0001	
Total_Length_of_Stay_Minutes	0.13863 <.0001	0.51486 <.0001	1.00000	0.26194 <.0001	
Number_of_Tests	-0.52734 <.0001	0.54630 <.0001	0.26194 <.0001	1.00000	

A Pearson correlation analysis was performed to assess the association between patient wait time, treatment time, length of stay, and number of laboratory tests. The analysis revealed that several variables had statistically significant relationships ($p < 0.0001$).

The two variables that showed the strongest positive correlation with each other were

Treatment_Time_Minutes and Number_of_Tests ($r = 0.5463$) (Figure 8), which implies that the higher the number of tests performed, the higher the treatment time tended to be. The second pair of variables with a strong positive correlation was Treatment_Time_Minutes and Total_Length_of_Stay_Minutes ($r = 0.5149$) (Figure 9), and thus, the length of treatment was associated with the length of a patient's stay.

Wait_Time_Minutes and Total_Length_of_Stay_Minutes had the lowest positive correlation ($r = 0.1386$), and the implication is that although the two variables are significantly associated, wait time has a minimal influence on the total length of stay. Other factors that influence the total length of stay have a more substantial impact.

The analysis revealed that there was a moderate negative correlation between Wait_Time_Minutes and Number_of_Tests ($r = -0.5273$). Thus, as the number of tests taken by the patient increased, the wait time decreased. In conclusion, it can be stated that patient treatment time, as well as the number of diagnostic tests, impact the total length of a patient's stay more significantly than the wait time.

Regression

Simple Linear Regression Model: Total Length of Stay Using Wait Time (Minutes)

The REG Procedure
Model: MODEL1
Dependent Variable: Total_Length_of_Stay_Minutes Total_Length_of_Stay_Minutes

Number of Observations Read	3000
Number of Observations Used	3000

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	93344	93344	58.75	<.0001
Error	2998	4763599	1588.92572		
Corrected Total	2999	4856943			

Root MSE	39.86133	R-Square	0.0192
Dependent Mean	188.85900	Adj R-Sq	0.0189
Coeff Var	21.10640		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	175.20079	1.92486	91.02	<.0001
Wait_Time_Minutes	Wait_Time_Minutes	1	0.31796	0.04148	7.66	<.0001

A simple linear regression analysis was performed to determine whether Wait_Time_Minutes could be used to predict the Total_Length_of_Stay_Minutes. The regression model that was estimated for this purpose was found to be significant, $F(1, 2998) = 58.75$, $p < 0.0001$, which means that there is a statistically significant association between the time spent waiting and the total length of the stay. In this regression model, the estimate for the regression coefficient was 0.318 ($t = 7.66$, $p < 0.0001$).

This can be interpreted as every one-minute increase in the waiting time is, on average, associated with the increase in the total length of stay by 0.32 minutes. The regression equation that can be used to predict the length of stay based on waiting time is as follows:

$$\text{Total Length of Stay} = 175.20 + 0.318 \times \text{Wait Time}$$

However, it should be noted that the total length of stay ($R^2 = 0.0192$), which suggests that only about 1.92% of the variance in the total length of stay can be explained by the waiting time. Thus, it can be concluded that although waiting time contributes to the increase in the total length of stay, it is a minor factor that can be influenced by other variables such as treatment time, the patient's condition, and the range of tests requested.

Two-Way ANOVA

Two-Way ANOVA: Department, Shift, and Interaction Effects on Wait Time

The GLM Procedure

Dependent Variable: Wait_Time_Minutes Wait_Time_Minutes

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	23	234999.7751	10217.3815	44.18	<.0001
Error	2976	688308.4169	231.2864		
Corrected Total	2999	923308.1920			

R-Square	Coeff Var	Root MSE	Wait_Time_Minutes Mean
0.254519	35.40391	15.20810	42.95600

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Department_ID	7	89120.5024	12731.5003	55.05	<.0001
Shift	2	134111.6250	67055.8125	289.93	<.0001
Department_ID*Shift	14	11767.6477	840.5463	3.63	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Department_ID	7	83341.12053	11905.87436	51.48	<.0001
Shift	2	94829.52802	47414.76401	205.00	<.0001
Department_ID*Shift	14	11767.64765	840.54626	3.63	<.0001

A two-way ANOVA was conducted to evaluate the effects of Department_ID and Shift on the patient wait time. In addition, the analysis tested if the effect of the shift was different for various departments.

Main effect: Department_ID

There was a significant main effect of the department on the wait time ($p < 0.0001$). Thus, it was found that in some departments, patients had to wait more on average than in others. The difference might be caused by the varying number of patients and staff in different units.

Main effect: Shift

There was a significant main effect of shift on the wait time ($p < 0.0001$). The shift affected the average waiting time; thus, patients arriving at different times of the day experienced different levels of healthcare service. It can be hypothesized that the number of patients varied across shifts, which led to changes in the patient load per staff.

Interaction effect: Department_ID $\times$ Shift

There was a significant interaction between Department_ID and Shift ($p < 0.0001$). The interaction implies that shifts affected waiting time differently depending on the department. Some units experienced a much bigger increase in their wait time than others during certain shifts ([Figure 10](#)).

Recommendations

1. Increase Staffing During Night Shifts

The analysis revealed that the Night shift has the largest average waiting time (52.17 minutes) with significantly higher waiting time durations than the Morning and Afternoon shifts. Therefore, the SmartHospital administration should consider increasing the number of doctors and equipment available for the Night shift to reduce waiting time.

2. Improve Efficiency in High-Demand Departments

Department level analysis showed that there are variances in terms of patient waiting time and volumes of patients in different departments. It is recommended to consider restructuring and

studying the processes within the most heavily loaded departments in order to provide a better organization of work and reduce the unnecessary waiting time of patients.

3. Reduce Treatment and Laboratory Delays

The multiple regression analysis found that Treatment_Time_Minutes and Lab_Delay_Minutes are the key factors in total length of stay. Enhancing treatment organization and reducing laboratory delay time might be beneficial in shortening the length of patients' stay.

4. Use Data Analytics for Continuous Monitoring

Hospital management should ensure that they frequently review and analyze performance measures such as time spent waiting, length of a hospital stay, workload by departments, and volume of patients. This is important for detecting problems before they become serious and addressing them.

5. Prioritize Patient Flow Improvement Initiatives

The analysis revealed that operational factors exert a more significant influence on the length of patient stay compared to demographic and physician-related aspects. Thus, it is recommended to optimize patient flow, eliminate treatment delays, and improve overall patient experience by standardizing and streamlining operations.

Limitations

There are a number of limitations in the study conducted. First, the data used was sourced from one hospital, which may affect the applicability of the results in the healthcare environment. Secondly, the study included variables associated with the waiting time of patients in the hospital. However, some of these variables could not be included because of their nature – for example, the number of nurses, patients, beds, or seasonal variability. Thirdly, some of the relationships discovered were weak; therefore, it is possible that there are multiple variables, which have an impact on the

performance of a hospital. Fourthly, the study utilized both historical analysis and observation as a method of research; which limits the ability to establish direct causal relationships between variables.

Conclusion

The research undertaken aimed to analyze the state of patient flow and service provisions in SmartHospital. Descriptive statistics, hypothesis tests, ANOVA, correlation analysis, regression analysis, and other statistical tools were used. The investigation revealed several vital facts about the operations of the facility in relation to patient flow. The first finding concerns the average waiting time, which was generally lower than the established 45-minute threshold. Still, the statistics varied significantly depending on the department and shift, with weekends and nightshifts experiencing higher delays. Secondly, the analysis found a disparity in the patient flow between different areas of the hospital. Further, the length of the stay was reported to be dependent on treatment, laboratory, and triage duration, as well as the patient's priority status. Overall, it was found that the use of analytics can help identify areas of improvement and support the staff in their efforts to enhance care. By optimizing patient flow, SmartHospital will be able to ensure higher efficiency and better care quality.

Appendix

Figure 1

HOSPITAL'S TRIAGE, DISPOSITION, DEPARTMENT SUMMARY:**The FREQ Procedure**

Department_ID				
Department_ID	Frequency	Percent	Cumulative Frequency	Cumulative Percent
D01	780	26.00	780	26.00
D02	328	10.93	1108	36.93
D03	334	11.13	1442	48.07
D04	280	9.33	1722	57.40
D05	515	17.17	2237	74.57
D06	315	10.50	2552	85.07
D07	254	8.47	2806	93.53
D08	194	6.47	3000	100.00

Disposition				
Disposition	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Admitted	1476	49.20	1476	49.20
Discharged	1315	43.83	2791	93.03
Transferred	209	6.97	3000	100.00

Triage_Level				
Triage_Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	241	8.03	241	8.03
2	536	17.87	777	25.90
3	1048	34.93	1825	60.83
4	840	28.00	2665	88.83
5	335	11.17	3000	100.00

Arrival_Mode				
Arrival_Mode	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Ambulance	790	26.33	790	26.33
Referral	426	14.20	1216	40.53
Walk-in	1784	59.47	3000	100.00

Figure 2

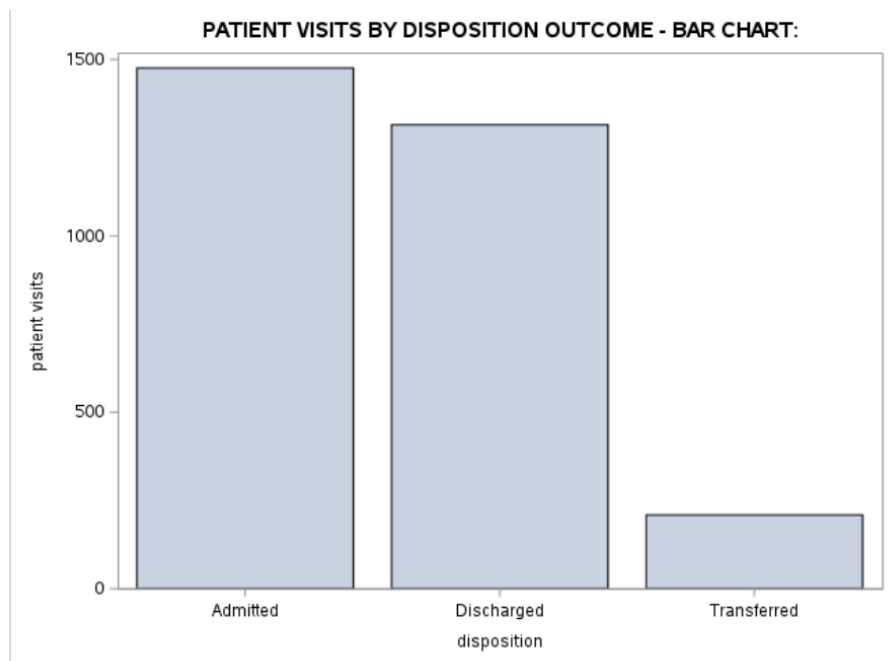


Figure 3

WAIT TIME VS LENGTH OF STAY - SCATTER PLOT:**The CORR Procedure**

1 With Variables:	Total_Length_of_Stay_Minutes
1 Variables:	Wait_Time_Minutes

Simple Statistics							
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
Total_Length_of_Stay_Minutes	3000	188.85900	40.24327	566577	86.00000	301.00000	Total_Length_of_Stay_Minutes
Wait_Time_Minutes	3000	42.95600	17.54628	128868	6.00000	98.00000	Wait_Time_Minutes

Pearson Correlation Coefficients, N = 3000 Prob > r under H0: Rho=0	
	Wait_Time_Minutes
Total_Length_of_Stay_Minutes	0.13863
Total_Length_of_Stay_Minutes	<.0001

Figure 4

AVERAGE WAIT TIME VS 45-MINUTE BENCHMARK- ONE SAMPLE TTEST:**The TTEST Procedure**

Variable: Wait_Time_Minutes (Wait_Time_Minutes)

N	Mean	Std Dev	Std Err	Minimum	Maximum
3000	42.9560	17.5463	0.3203	6.0000	98.0000

Mean	95% CL Mean	Std Dev	95% CL Std Dev
42.9560	42.3279 43.5841	17.5463	17.1133 18.0019

DF	t Value	Pr > t
2999	-6.38	<.0001

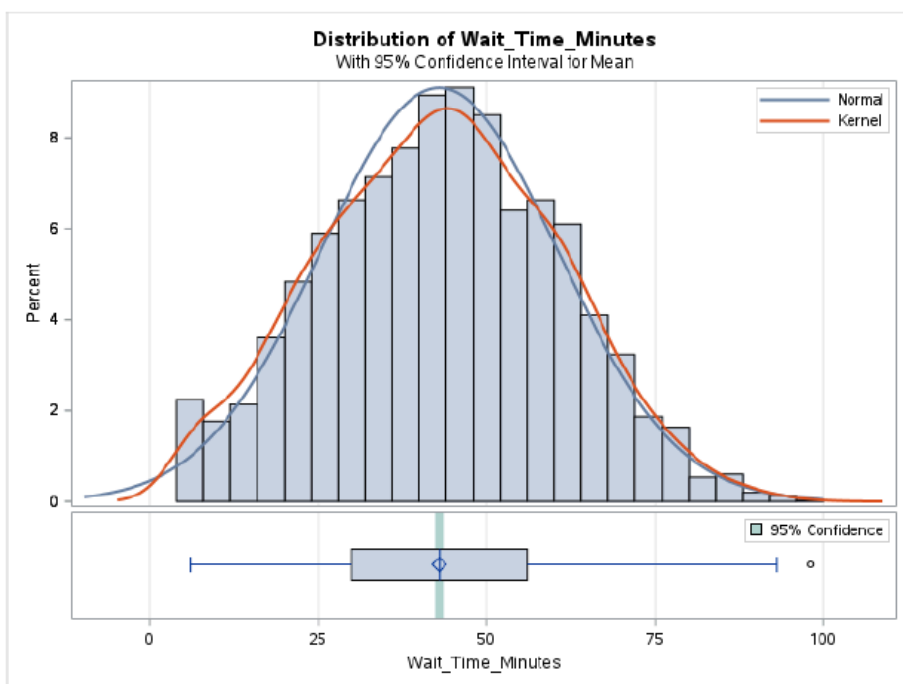


Figure 5

COMPARING WAIT TIME FOR WEEKDAY AND WEEKEND VISITS - INDEPENDENT TWO SAMPLE TTEST:

The TTEST Procedure

Variable: Wait_Time_Minutes (Wait_Time_Minutes)

Weekend_Flag	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		2094	41.4422	17.4192	0.3807	6.0000	93.0000
1		906	46.4547	17.3473	0.5763	6.0000	98.0000
Diff (1-2)	Pooled		-5.0125	17.3976	0.6918		
Diff (1-2)	Satterthwaite		-5.0125		0.6907		

Weekend_Flag	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		41.4422	40.6957	42.1887	17.4192	16.9072	17.9635
1		46.4547	45.3237	47.5858	17.3473	16.5837	18.1852
Diff (1-2)	Pooled	-5.0125	-6.3690	-3.6560	17.3976	16.9682	17.8494
Diff (1-2)	Satterthwaite	-5.0125	-6.3672	-3.6578			

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	2998	-7.25	<.0001
Satterthwaite	Unequal	1724.9	-7.26	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	2093	905	1.01	0.8890

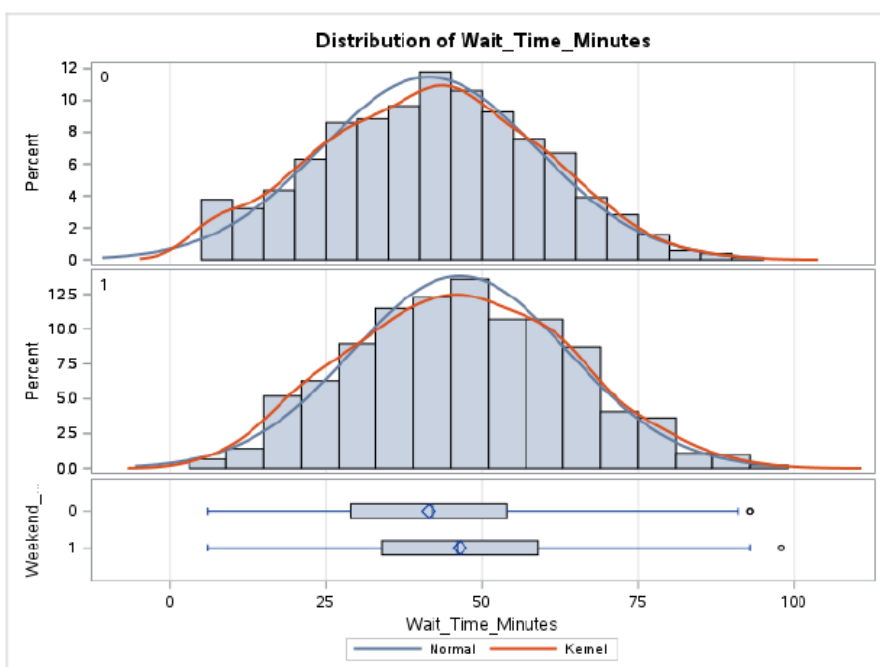


Figure 6

One-Way ANOVA: Patient Wait Time by Departments (Minutes)		
The ANOVA Procedure		
Class Level Information		
Class	Levels	Values
Department_Name	8	Cardiology Emergency Medicine Gastroenterology General Medicine Neurology Orthopedics Pediatrics Respiratory Medicine
Number of Observations Read		3000
Number of Observations Used		3000

One-Way ANOVA: Patient Wait Time by Departments (Minutes)

The ANOVA Procedure

Dependent Variable: Wait_Time_Minutes Wait_Time_Minutes

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	89120.5024	12731.5003	45.66	<.0001
Error	2992	834187.6896	278.8060		
Corrected Total	2999	923308.1920			

R-Square	Coeff Var	Root MSE	Wait_Time_Minutes Mean
0.096523	38.87114	16.69749	42.95600

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Department_Name	7	89120.50237	12731.50034	45.66	<.0001

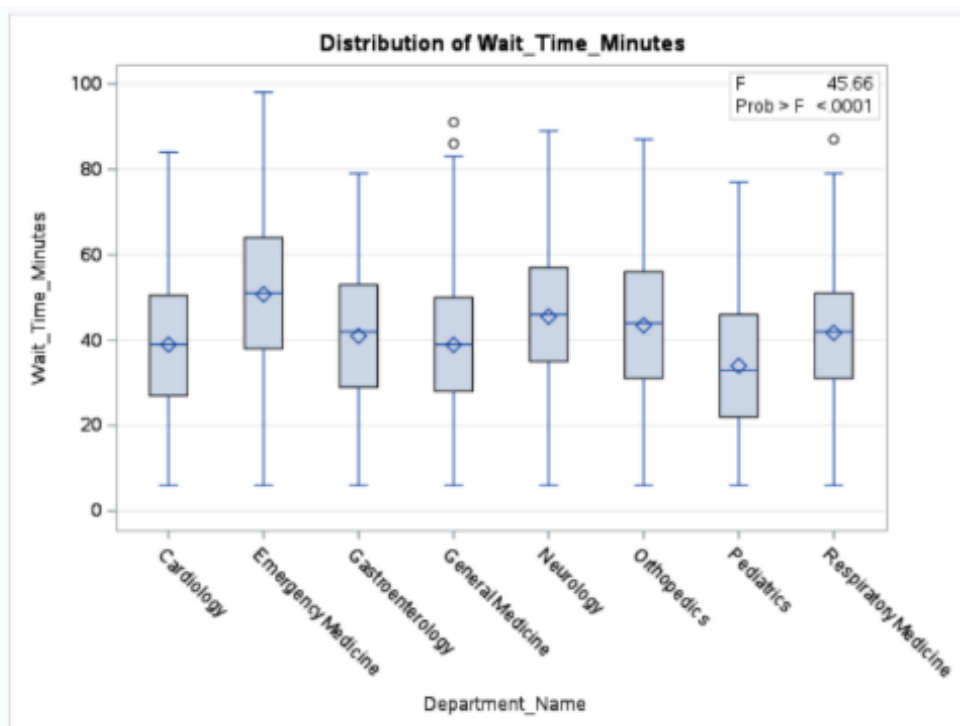


Figure 7

Tukey Post Hoc Test: Department Differences in Wait Time (Minutes)

The ANOVA Procedure

Tukey's Studentized Range (HSD) Test for Wait_Time_Minutes

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	2992
Error Mean Square	278.806
Critical Value of Studentized Range	4.28936

Tukey Post Hoc Test: Pairwise Shift Comparisons for Wait Time

The GLM Procedure

Least Squares Means

Adjustment for Multiple Comparisons: Tukey-Kramer

Shift	Wait_Time_Minutes LSMEAN	LSMEAN Number
Afternoon	43.9835861	1
Morning	34.7533516	2
Night	50.1771122	3

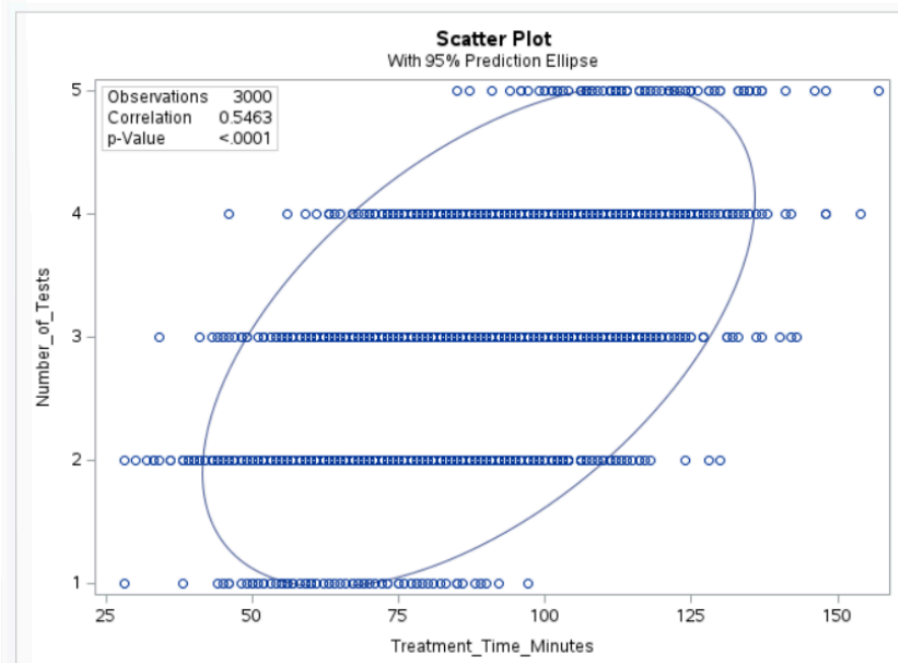
Least Squares Means for effect Shift Pr > t for H0: LSMean(i)=LSMean(j)			
Dependent Variable: Wait_Time_Minutes			
i/j	1	2	3
1		<.0001	<.0001
2	<.0001		<.0001
3	<.0001	<.0001	

Shift	Wait_Time_Minutes LSMEAN	95% Confidence Limits	
Afternoon	43.983586	43.015610	44.951562
Morning	34.753352	33.818494	35.688209
Night	50.177112	48.925440	51.428785

Least Squares Means for Effect Shift			
i	j	Difference Between Means	Simultaneous 95% Confidence Limits for LSMean(i)-LSMean(j)
1	2	9.230234	7.620899 10.839570
1	3	-6.193526	-8.085796 -4.301256
2	3	-15.423761	-17.292064 -13.555458

Figure 8

Treatment time vs number of tests:



Wait time vs number of tests:

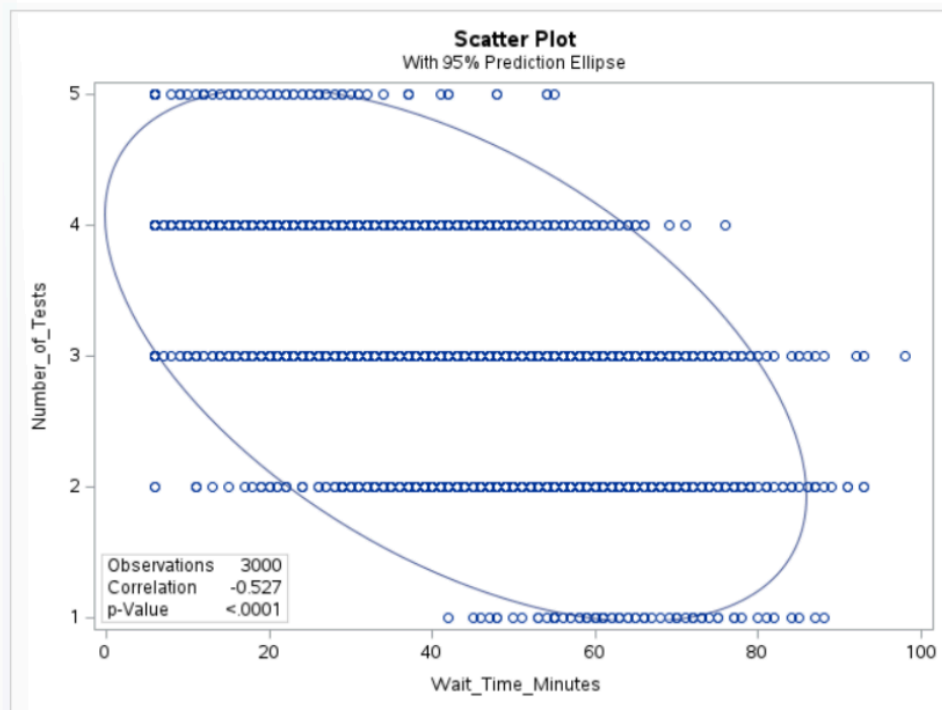
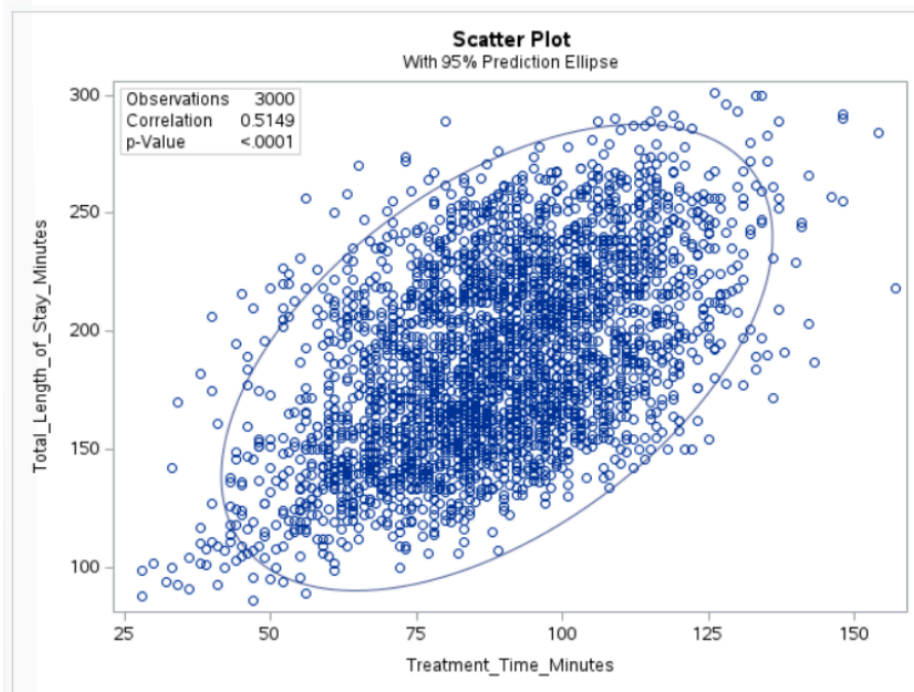


Figure 9

Treatment time vs total length of stay:



Wait time vs total length of stay:

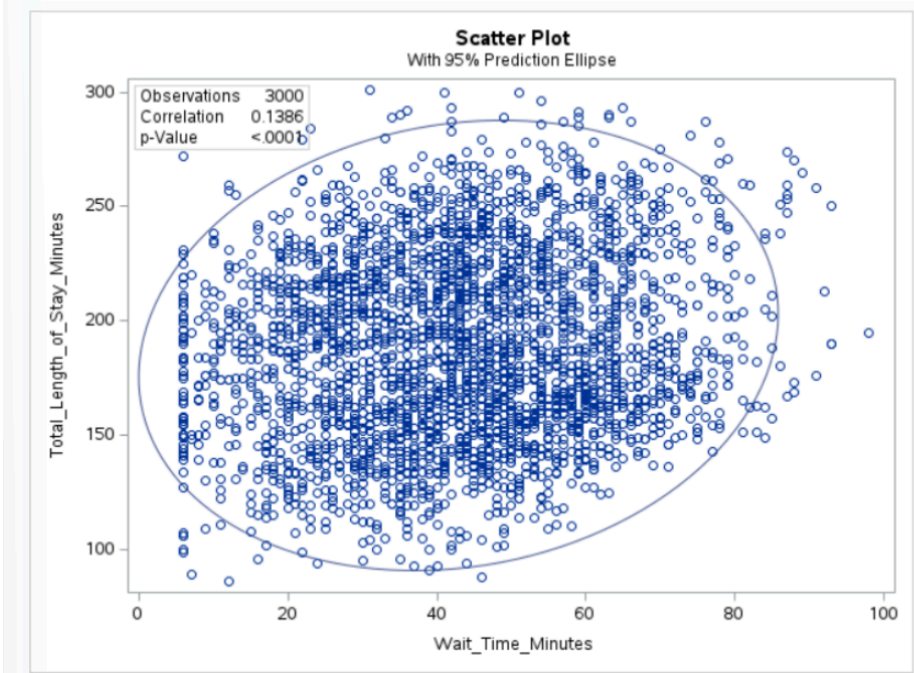


Figure 10

